

COURSE SYLLABUS
IST 615 CLOUD MANAGEMENT
(IST 615-M001 48507)

Instructor: Jim Garrisi

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Office Hours: TBA

Term: Spring 2026

Course Time: Tuesdays 5:00pm – 7:45pm

Classroom: Hinds 010

Course Description:

This course will equip students with skills in cloud enterprise service creation and management, and technical and business knowledge required for assessing opportunities and risks of cloud services.

Additional Course Description:

The course focuses on developing skills to manage cloud instances through lab assignments. Some of the labs use the services/infrastructure provided by top Cloud vendors in the market today. We note the course uses – cloud services – but some activities must be done in the student's own accounts and machines. Conceptual themes will also be presented alongside technical aspects of cloud management.

Prerequisite / Co-requisite:

Basic knowledge or computer networking concepts and the operation of the TCP/IP protocol suite is needed. This may be acquired through courses or online resources

Audience:

Graduate students who are interested to learn about cloud services and their management.

Credits:

3

Learning Objectives:

After taking this course, the students will be able to:

- Manage cloud services:
 - Make strategic choices for cloud digital service creation and monetization across different platforms
 - Assess accurately how cloud management approaches, technology, policy and governance may affect human users, devices/other non-person entities, and individual organizations

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- Design, configure, and manage hybrid, and on-premises cloud-based information services in digital enterprise environments.
- Critically assess cloud services
 - Interpret and analyze key factors to be considered before moving workloads to a cloud environment
 - Identify the necessary security and policy requirements needed for deploying a cloud-service
- Create a cloud-based application
 - Use services provided by vendors to host applications in cloud environments
 - Identify the optimal cloud service platform to host an application in order to minimize the cost and maximize its performance

Texts / Supplies:

This course will make use of content from several books but no specific book is required. Additional readings in PDF format or from websites will be posted in Blackboard by the instructor.

Throughout the course different cloud services will be used and some activities must be done in the student's own accounts and machines. In particular, **students will be strongly advised to obtain a special subscription to Microsoft Azure cloud services** that has been setup through a collaboration between Microsoft and the iSchool. More details on how to acquire the subscription through the University's bookstore will be provided in the first class sessions.

Course Requirements and Expectations:

- Lab and Homework assignments will be used to re-enforce and evaluate understanding and development of concepts covered in one or two weeks of content.
- All of the laboratory exercises will be performed outside of class hours.

General Requirement for Assignment Submissions

Assignments must be professionally prepared with computer-based applications. Unless otherwise stated, assignments must be submitted electronically to Blackboard. No hand-written assignments will be accepted. An assignment must be in one document when it is submitted to Blackboard. If you have additional supporting materials that are in physical forms or hard copies (e.g., business forms or some images), you must scan them into JPG or TIFF format and embed them into your MS Word or PDF document. If you use MS Word to prepare your assignments, use Times New Roman style with 12-pt font, 1.5 line spacing, and 1 inch margin around.

For each assignment, be sure to include a title page with the following information:

- Course number (i.e. IST 615)
- Name(s) of the author(s) or the team members

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- Name of the assignment (e.g. "Homework 2")
- Date the assignment is due and date in which it is submitted
- Number pages in the body of the assignment (not necessary on the first page)

You must submit all assignments to the professor in class on the deadline specified for each assignment. If you are late, I will deduct 10% of the original grade for the first day of lateness plus 15% for every subsequent day. You will not receive full credit for topics/assignments presented in an unprofessional manner. Professionalism includes the proper use of grammar, punctuation, and limiting spelling mistakes. Professionalism also means adhering to given instructions. Failure to adhere to the assignment instructions will result in a reduction of the grade. If English is not your first language set up an appointment with the writing program so they can help you improve your writing.

How to Succeed in This Course:

Keep pace with the content of the course. It is expected that you will spend 6 to 9 hours outside of class time covering the material, the readings and executing labs / homework and project assignments.

Mental health and overall well-being are significant predictors of academic success. As such it is essential that during your college experience you develop the skills and resources to effectively navigate stress, anxiety, depression and other mental health concerns. Please familiarize yourself with the range of resources the Barnes Center provides (<https://ese.syr.edu/bewell/>) and seek out support for mental health concerns as needed. Counseling services are available 24/7, 365 days, at 315-443-8000. I encourage you to explore the resources available through the Wellness Leadership Institute, <https://ese.syr.edu/bewell/wellness-leadership-institute/>.

Grading:

All exams and assignments will be graded in a 0-100 point scale. The weight distribution of the different activities/assignments that you will work on in this course is mentioned in the following table.

Activity / Assignment	Final grade percentage value
Attendance and Participation in Class	10%
Laboratory exercises and Homework	50%
Midterm exam	18%
Final exam	22%

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Grading Table:

The following table will be used in the assignment of your final letter grade for the course.

Letter Grade	Grade points /credit	Percentage Range	Total Points
A	4.000	93% - 100%	93 – 100
A-	3.667	90% - 92.99%	90 – 92.99
B+	3.333	87% - 89.99%	87 – 89.99
B	3.000	83% - 86.99%	83 – 86.99
B-	2.667	80% - 82.99%	80 – 82.99
C+	2.333	77% - 79.99%	77 – 79.99
C	2.000	73% - 76.99%	73 – 76.99
C-	1.6667	70% - 72.99%	70 – 72.99
F	0	0% - 69.99%	0 – 69.99

Syracuse University Policies:

Syracuse University has a variety of policies designed to guarantee that students live and study in a community respectful of their needs and those of fellow students. **The policies and services are listed on the new Syracuse University Senate approved syllabus appendix titled, ‘Syracuse University Student Policies and Services’.** These statements are an official part of this course syllabus.

Course specific policies:

Attendance

In addition to the university’s attendance policy, please note that attendance will be taken throughout the semester in both labs and lectures. If you arrive late or leave early, you will be marked absent. There are no excused absences unless documented by the university. If you have 2 or more absences, your final course grade will be dropped one level down the grade scale. (A- is lowered to B+, C+ becomes a C, etc.)

Use of Class Materials and Recordings:

Original class materials (handouts, assignments, tests, etc.) and recordings of class sessions are the intellectual property of the course instructor. You may download these materials for your use in this class. However, you may not provide these materials to other parties (e.g., web sites, social media, other students) without permission. Doing so is a violation of intellectual property law and of the student code of conduct.

Educational Use of Student Work

Student work prepared for University courses in any media may be used for educational purposes, if the course syllabus makes clear that such use may occur. You grant permission to have your work used in this manner by registering for, and by continuing to be enrolled in, courses where such use of student work is announced in the course syllabus.

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I intend to use academic work that you complete this semester in subsequent semesters for educational purposes. Before using your work for that purpose, I will either get your written permission or render the work anonymous by removing all your personal identification.

Course Schedule:

The following table shows the breakdown of topics to be covered in this course. This table is **subject to modification as the course evolves**. Up to date information (especially for homework assignments) will be posted in Blackboard.

Session #	Date	Topic
1	1/13/26	Introduction: Compute Network and Storage IAAS, PAAS, SAAS
2	1/20/26	Cloud and Networking Technology Fundamentals 1 - Virtualization - Containers - Serverless computing
3	1/27/26	Cloud and Networking Technology Fundamentals 2 - TCP/IP Introduction to Azure - Using Azure - Example
4	2/3/26	Cloud and Networking Technology Fundamentals3 Introduction to AWS
5	2/10/26	Cloud Capacity Management AWS and Azure: SLAs and compute resources
6	2/17/26	Containers and Kubernetes
7	2/24/26	No Class: Midterm Review
8	3/3/26	Midterm
	3/10/26	Spring Break
9	3/17/26	Serverless computing Microservices
10	3/24/26	Microservices 2
11	3/31/26	Cloud security
12	4/7/26	DevOps and Automation
13	4/14/26	Hybrid Clouds & Edge Computing Cloud technology trends FinOps
14	4/21/26	Final Review

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15	4/28/26	Final Exam (in class)
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